Title: Leveraging Automation for Adaptive Planning in MRI-Guided Prostate Cancer Focal Cryoablation

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Categories (category - subcategory)
Interventional - MR-Guided Interventions
AI and Machine Learning - AI/ML Software

Synopsis (100 words):
Motivation: Predicting freezing volume intraoperatively is essential for effective focal cryoablation of prostate cancer. The lack of a fast, accurate, and clinically viable method currently hinders the development of robust planning software for optimal cryo-needle placement.
Goal: To develop an automated method to predict final iceball formation as cryo-needles are inserted.
Approach: We use automated segmentation of needles and urethra from MRI images, which serves as input for a predictive model to estimate iceball formation.
Results: Tested on five retrospective cases, the software demonstrated feasibility, enabling rapid, one-click predictions within seconds.

Impact (40 words)
Integrating automatic AI-based segmentation for rapid iceball prediction enables adaptive, real-time planning in cryoablation. This approach minimizes labor-intensive segmentation, allowing updates with each needle insertion to ensure comprehensive tumor ablation, enhancing precision, confidence, and efficiency in prostate cancer treatment.

Main body (750 words)
Introduction

Focal cryoablation is only effective if the iceball (frozen tissue volume) adequately covers the target volume while sparing critical structure[1,2]. Estimating the iceball is significantly challenging. Multiple factors, including the distance between needles, the urethral warmer catheter, and prostate boundaries, influence the formation of iceballs. Currently, physicians cognitively estimate iceball size using vendor guidelines and personal expertise, which can lead to insufficient coverage or unintended damage to surrounding tissues, increasing the risk of recurrence. Previous studies on computer-assisted planning tools for prostate cryoablation have primarily focused on pre-operative planning of whole-gland or hemi-gland ablation. The available methods rely on either finite element method or bioheat simulation, and demand a substantial amount of computational time, limiting its use intraoperatively. Nevertheless, various factors hinder the procedure planning days before the procedure, such as prostate displacement caused by the insertion of the urethral catheter, local anesthesia, and differences in patient positioning between the diagnostic MRI and the procedure position. Therefore, procedure planning occurs intraoperatively and is inherently iterative, as the physician may adjust the treatment plan after each needle insertion. In this study we present an

autonomous software based on deep learning algorithms to estimate the iceball formation intraoperatively after the cryo-needles are placed and before the freezing protocol starts.

Methods

Our approach to predicting iceball formation in prostate cryoablation consists of two primary steps: automated segmentation of needles and urethra from MRI images, followed by predictive modeling to estimate iceball formation based on these segmentations. The models were trained and validated using a retrospective dataset of 38 intraoperative MRIs collected during prostate cancer cryoablation procedures. We excluded patients who had previously undergone prostatectomy or whole-gland cryoablation. The segmentation and prediction were performed in the confirmation image performed after the needles are inserted (T2-weighted turbo spin echo sequence, TR/TE=2430-3754/104-106ms, matrix=192-320×134-256, number of slices=18-30, slice thickness=3-3.6mm, FOV=180-192mm×157-240mm).

In the first step, the urethral warming catheter and needles are segmented using DiNTS[3] for the urethra and SwinUNETR[4] for the needles. These segmentations are refined with dilated prostate segmentations to ensure both are within the prostate, which is also segmented using SwinUNETR. Finally, for the iceball prediction model, the SegResNet model was used which was trained over 100 epochs using DiceCELoss as the loss function. This pipeline was then integrated into 3D Slicer by developing a custom extension based on the MONAI Auto3DSeg. The entire process is performed automatically, following the flowchart described in Fig. 1.

Retrospective Analysis: We conducted a comparative evaluation of our fully autonomous iceball prediction model against the semi-autonomous approach presented by Moreira et al[5] in five cases that were not used for training. In our framework, user interaction is limited to selecting the 'start' button, whereas Moreira et al.'s interface required manual segmentation of the urethra and localization of needle. Therefore, we analyzed: (1) *Intraoperative usability assessment*. To evaluate the model's feasibility for intraoperative use, we measured the number of user interactions (mouse clicks) required and the time needed to generate a predicted iceball once MRI images were loaded into 3D Slicer. (2) *Prediction Accuracy*. The accuracy of each prediction was determined using the Dice Similarity Coefficient (DSC) against the manually segmented iceball at the end of the first freezing cycle.

Results

(1) *Interoperability Feasibility Analysis*. Using our autonomous approach only one click was required in all cases, and final iceball prediction took a total time of 30.2±2.8s. On the other hand, using the semi-autonomous, it took an average of 24.8 ± 3.0 clicks to produce an iceball prediction in an average of 58.8±13.8s.

(2) *Prediction Accuracy.* Both approaches offered a similar accuracy, with the fully autonomous and semi-autonomous approaches providing an average DSC of 0.65±0.7 and 0.63±0.8, respectively.

Discussion

The results indicate that the fully autonomous approach runs in approximately half the time required by the semi-autonomous one, while keeping the same level of accuracy. A representative case is presented in Fig. 2. Our result suggests that the fully autonomous model is well-suited for intraoperative use, where speed and precision are critical. The streamlined

performance of the autonomous model contrasts with the semi-autonomous method, where prediction outcomes may vary due to user-dependent segmentation.

Conclusion

This study highlights the potential of integrating AI-based segmentation tools to achieve rapid and precise iceball predictions, effectively eliminating the labor-intensive task of intraoperative structure segmentation. Our findings suggest that this approach enables adaptive intraoperative planning, where iceball predictions can be updated in real time following each needle insertion, ensuring comprehensive tumor ablation with greater confidence and efficiency.

Acknowledgments

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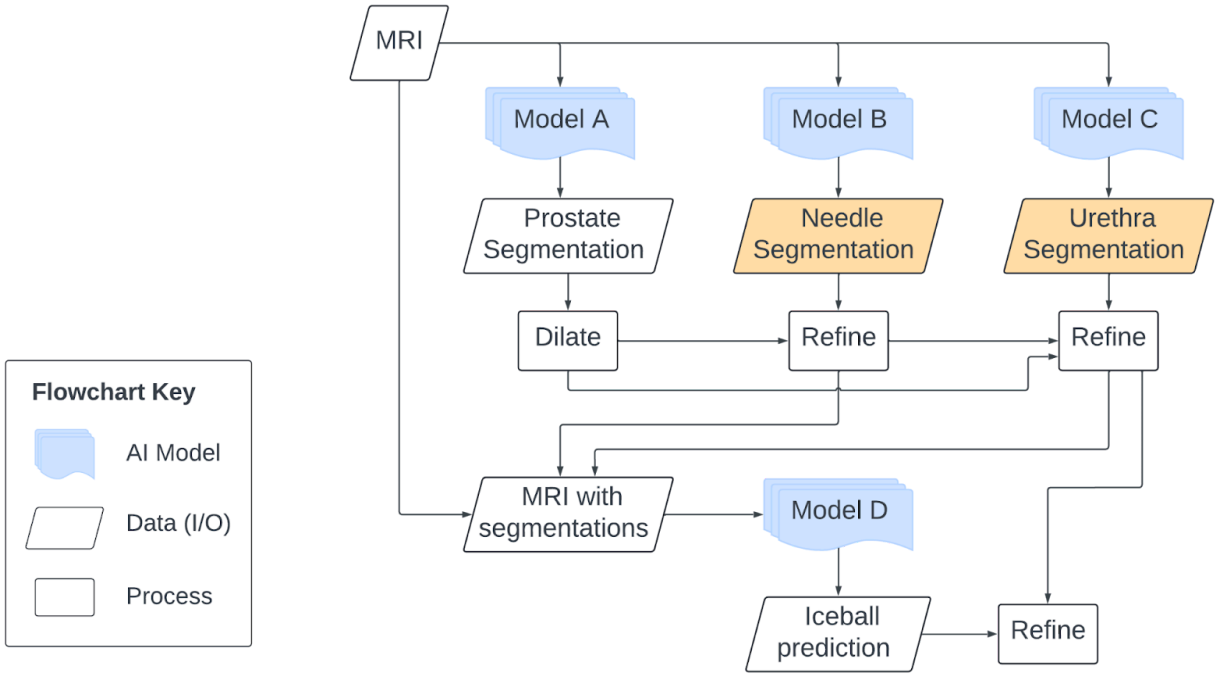


Fig. 1 Workflow of autonomous iceball prediction model. Three AI-based models (models A,B, and C) are used to prepare the input of the iceball prediction model (model D).

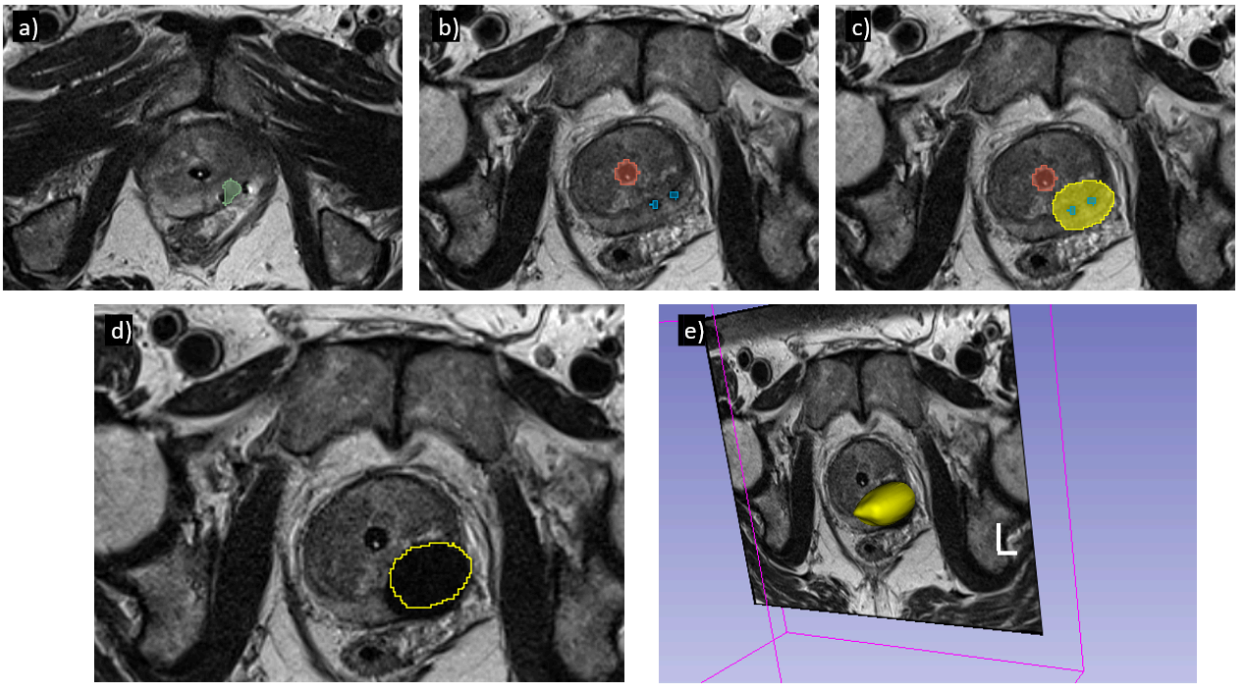
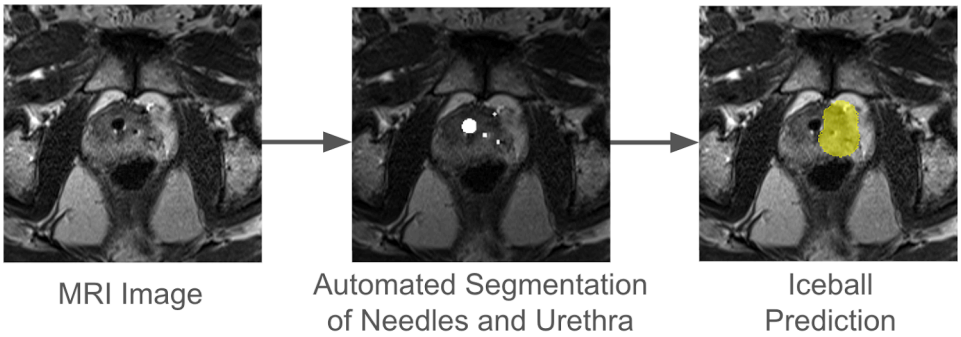


Fig. 2 Representative results of focal cryoablation in a 70-year-old patient with a small lesion in the left peripheral zone. (a) Initial imaging showing the lesion. (b) Automatic segmentation of the urethral catheter and cryo-needles. (c) Iceball prediction generated by the prediction model. (d)

Slice view and (e) 3D view of the predicted (yellow) and achieved iceball (darker region), demonstrating the outcome at the end of the first freezing cycle.



Preview figure